

Multiple Choice (10 marks)

1. Let $x = 1$. What is $x \ll 3$ in Python 3?

- (a) 1
- (b) 3
- (c) 8
- (d) 16

2. Which is the largest asymptotically?

- (a) $O(1)$
- (b) $O(n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

3. Which of the following statements about a remote procedure call is true?

- (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
- (b) It cannot return a value.
- (c) It cannot pass parameters by reference.
- (d) It cannot call procedures implemented in a different language.

4. What is the output of `"abc"[:-1]` in Python 3?

- (a) Error
- (b) abc
- (c) cba
- (d) c

5. In Python 3, what is the output of `['Hi!'] * 4`?

`'Hi!', 'Hi!', 'Hi!', 'Hi!'`

`'Hi!' * 4`

- (a) Error
- (b) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, converting a list with duplicate elements to a set results in a collection of unique elements.
7. True or False: Identifier length is best defined using a context-free grammar in programming language specifications.
8. True or False: Choosing to busy-wait on an asynchronous event is unreasonable in a time-sharing system due to potential resource wastage.
9. True or False: In Python 3, the `join()` function checks if all characters in a string are uppercase.
10. True or False: In Python 3, the operator `'//'` is used to perform exponential (power) calculations on operands.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that gives profit amount if the given amount has profit else return none.
12. Write a function to find the maximum sum possible by using the given equation $f(n) = \max((f(n/2) + f(n/3) + f(n/4) + f(n/5)), n)$.

End of Paper